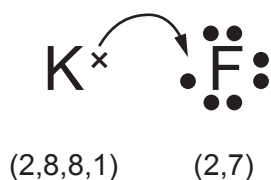


Examiner
only

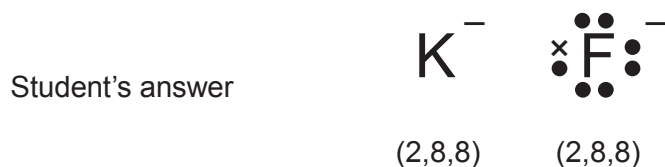
- 8.** (a) The table shows the electronic structure of the elements present in potassium fluoride.

Element	Electronic structure
potassium	2,8,8,1
fluorine	2,7

The diagram shows the electron transfer that occurs when potassium reacts with fluorine to form potassium fluoride. The ● and × symbols are outer shell electrons.



A student was asked to draw a diagram showing the electronic structures and charges on the ions formed. There are **two** mistakes in the student's answer.

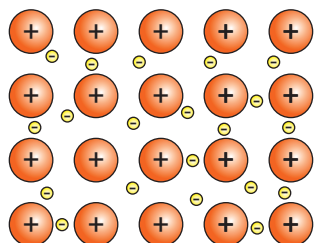


- (i) Circle the **two** mistakes in the student's answer. [2]

- (ii) Name the type of bonding found in potassium fluoride. [1]



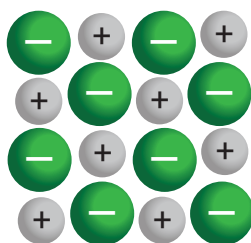
- (iii) The diagrams show four different structures. Give the **letter** of the structure most likely to represent potassium fluoride. [1]



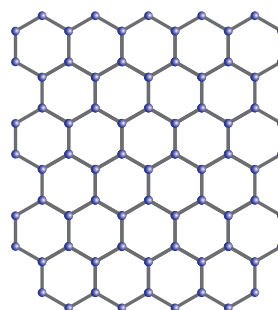
A



B



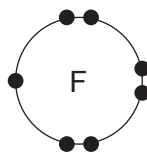
C



D

Letter

- (b) The diagram shows the electrons in the outer shell of an atom of fluorine.



Complete the diagram to show the outer shell electrons in a molecule of fluorine. [2]

